

MATH 351 Midterm 2

Fall 2025

Name _____

Rules of the test

- Time limit: test is due at the end of our usual class time. I have written the test expecting you to finish within 85 minutes.
- There are four problems. I will weight them equally.
- All four problems have multiple parts. You may be able to complete later parts successfully even if you have not completed earlier parts successfully.
- Read each problem carefully. Many details in what I have written are important to the interpretation of the problems. Give clear explanations using the context of the problem.
- If you are uncertain about what to do with any of the questions on the test, please ask me so that I can point you in the right direction.
- The exam is to be taken individually—no collaboration of any kind is allowed.
- Here is the allowed aid for this test. You may use a single-sided handwritten sheet of notes during the test. You will submit your sheet of notes with your completed exam. You may also use a calculator on all questions. You may not use your phone as a calculator. If you need to borrow a calculator, I can provide one.
- If you need to leave the room during the exam, you must leave your phone with me.
- If I have reason to suspect inappropriate collaboration or use of aid that I have not allowed for this test, I will give everyone involved a score of 0.

Pledge

When you are done, come back here to sign the honor pledge.

I pledge that I have neither given nor received any unauthorized aid on this exam.

Signed: _____

1. I will play a simple game with a deck of cards. I draw a card at random from the deck, hoping to get a heart card. If I do not get a heart, I draw a second card from the deck. I keep my first card; it does not go back in the deck. I win, if I get heart card on either the first or second draw.

a. What is the probability that I get a heart on the first draw?

Let A be the event that I get a heart on the first draw. We have $P(A) = 13/52 = 1/4$.

b. Assuming that I do not get a heart on the first draw, what is the probability that I get a heart on the second draw?

This conditional probability is $13/51$.

c. What is the probability that I win the game?

Let B be the event that I win the game. We calculate $P(B)$ by thinking of the game as a two-stage experiment.

$$P(B) = P(A)P(B \mid A) + P(A^c)P(B \mid A^c) = (1/4)(1) + (3/4)(13/51) = 1/4(1 + 13/17) = 30/68 = 15/34.$$

I calculated $P(B \mid A^c)$ as $13/51$ by “reducing the sample space.” Given that I didn’t win (get a heart) on the first draw (conditioning on the occurrence of A^c), to win the game, I need to get a heart on the second draw. The (conditional) probability of that event is $13/51$.

d. I played this game and I won. Given that information, what is the probability that I did **not** get a heart on the first draw?

We calculate $P(A^c \mid B)$ using the Bayes’ formula method:

$$P(A^c \mid B) = P(B \mid A^c)P(A^c) / P(B) = (13/51)(3/4) / (15/34) = (13/68) / (15/34) = 13/30.$$

2. Suppose you roll two *four-sided* dice marked with numbers 1 through 4. Let X be the random variable representing the higher of the two numbers rolled, and let Y be the random variable representing the lower of the two numbers rolled. For example, if you roll a 2 and 4, then X is 4 and Y is 2. If, on the other hand, you roll a 3 and a 3, then X is 3 and Y is 3.

- a. Fill in the table below with the joint distribution for X and Y . (I filled in one slot for you.)

	$X = 1$	$X = 2$	$X = 3$	$X = 4$	
$Y = 1$	$1/16$	$2/16$	$2/16$	$2/16$	$7/16$
$Y = 2$	0	$1/16$	$2/16$	$2/16$	$5/16$
$Y = 3$	0	0	$1/16$	$2/16$	$3/16$
$Y = 4$	0	0	0	$1/16$	$1/16$
	$1/16$	$3/16$	$5/16$	$7/16$	

- b. Add the marginal distributions for X and Y to your table, labeling the marginals appropriately.
- c. Are X and Y independent? Explain your answer briefly and precisely using the table.

They are not independent. For example, we have $P(X = 1 \cap Y = 1) = 1/16$ but $P(Y=1)P(X=1) = (7/16)(1/16) = 7/256$.

- d. Given that the higher roll is a 4, what is the expected value of the lower roll? Please show your work and give your result as an exact value—a fraction—rather than as a decimal approximation.

The conditional distribution for Y is the following:

$$P(Y = 1 \mid X = 4) = 2/7; P(Y = 2 \mid X = 4) = 2/7; P(Y = 3 \mid X = 4) = 2/7; P(Y = 4 \mid X = 4) = 1/7.$$

$$\text{We thus have } E[Y \mid X = 4] = (1)(2/7) + (2)(2/7) + (3)(2/7) + (4)(1/7) = (2 + 4 + 6 + 4) / 7 = 16 / 7.$$

3. The random variable X has density function f defined by $f(x)=0$, if $x<0$; $f(x)=20x^3(1-x)$, if $0\leq x\leq 1$; and $f(x)=0$, if $x>1$.
- a. Calculate the distribution function F for X . Your formula for F should include cases for all input values and not just for inputs x between 0 and 1.

We can calculate the distribution function as follows:

- For $x<0$, we have $F(x)=0$.
- For $x>1$, we have $F(x)=1$.
- For $0\leq x\leq 1$, we have $F(x)=\int_0^x f(t)dt=\int_0^x 20t^3(1-t)dt=\int_0^x (20t^3-20t^4)dt=5x^4-4x^5$.

- b. Calculate $P\left(\frac{1}{3}\leq X\leq\frac{2}{3}\right)$.

We can calculate $P\left(\frac{1}{3}\leq X\leq\frac{2}{3}\right)$ using the distribution function.

$$P\left(\frac{1}{3}\leq X\leq\frac{2}{3}\right)=F\left(\frac{2}{3}\right)-F\left(\frac{1}{3}\right)=\left(5\left(\frac{2}{3}\right)^4-4\left(\frac{2}{3}\right)^5\right)-\left(5\left(\frac{1}{3}\right)^4-4\left(\frac{1}{3}\right)^5\right).$$

We can leave the result in this form.

- c. Calculate $E[X]$.

$$E[X]=\int_0^1 xf(x)dx=\int_0^1 20x^4(1-x)dx=\int_0^1 (20x^4-20x^5)dx=\frac{20}{5}-\frac{20}{6}=4-\frac{10}{3}=\frac{2}{3}$$

- d. Calculate $\text{Var}(X)$.

$$\text{Var}(X)=E[X^2]-E[X]^2$$

$$E[X^2]=\int_0^1 x^2 \cdot 20x^3(1-x)dx=\int_0^1 (20x^5-20x^6)dx=\frac{20}{6}-\frac{20}{7}=\frac{20}{42}=\frac{10}{21}$$

$$\text{Var}(X)=\frac{10}{21}-\left(\frac{2}{3}\right)^2=\frac{10}{21}-\frac{4}{9}=\frac{30-28}{63}=\frac{2}{63}.$$

4. When my children were small, my mother-in-law, Theresa, would call every week to talk to them. We can model the length of her weekly telephone calls as independent random variables all having the same exponential(λ) distribution.

a. Theresa's average call was 10 minutes. Based on this piece of information, what value would you use for λ ?

If $X \sim \text{exponential}(\lambda)$, then $E[X] = 1/\lambda$. We match $1/\lambda$ with 10 to get $\lambda = 1/10$.

b. According to your model, what is the probability that one of Theresa's weekly phone calls lasts at least 20 minutes? (Your value should be around 0.1.)

The distribution function F for X is given by $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$. We need

$$P(X \geq 20) = 1 - P(X < 20) = 1 - (1 - e^{-\lambda 20}) = e^{-\frac{1}{10} 20} = e^{-2} \approx 0.1353.$$

c. Let N be the random variable representing the number of long calls (20 minutes or longer) that we get in a year (52 weeks). What distribution would you use for N ?

We would use $N \sim \text{binomial}(52, e^{-2})$, since the weekly call lengths are independent.

d. What are the expected value and variance of N ?

$$E[N] = 52(e^{-2}) \approx 7 \text{ long calls}$$

$$\text{Var}(N) = 52(e^{-2})(1 - e^{-2}) \approx 6.085 \text{ (which gives standard deviation around 2.5 calls)}$$

e. Let Y be the length of the longest phone call in a month (say, four weekends). What is the distribution function for Y according to our model? In other words, give a formula for $P(Y \leq x)$.

There are four calls in the month. Let their lengths be X_1, X_2, X_3, X_4 .

$$P(Y \leq x) = P((X_1 \leq x) \cap (X_2 \leq x) \cap (X_3 \leq x) \cap (X_4 \leq x)) = P(X_1 \leq x)P(X_2 \leq x)P(X_3 \leq x)P(X_4 \leq x),$$

using the independence of the weekly call lengths.

Since the X_i all have the exponential(λ) distribution, this probability is $(1 - e^{-\lambda x})^4$.